Experiment 3

Date : 31-01-2026

LINEAR DISCRIMINANT ANALYSIS

Description :

Linear Discriminant Analysis

is generally used in supervised learning algorithms as a classifier to separate

the input data into two or more classes. It is also a dimensionality reduction

technique.

Algorithm :

Step 1: Compute

the d-dimensional mean vectors for the different classes from the dataset.\

Step 2: Compute the scatter

matrices (in-between-class and within-class scatter matrix).

Step 3: Compute the

eigenvectors and corresponding eigenvalues for the scatter matrices.

Step 4: Sort the eigenvectors

by decreasing eigenvalues and choose eigenvectors with the largest

eigenvalues to form a matrix (where every column represents an eigenvector).

Step 5: Use this

eigenvector matrix to transform the samples onto the new subspace.

Dataset :
https://archive.ics.uci.edu/ml/machine-learning-databases/balance-scale/balance-scale.data

Dataset Information :

This data set was generated to model psychological experimental results. Each example is classified as having the balance scale tip to the right, tip to the left, or be balanced. The attributes are the left weight, the left distance, the right weight, and the right distance. The correct way to find the class is the greater of (left-distance * left-weight) and (right-distance * right-weight). If they are equal, it is balanced.

No.of instances: 625

No.of attributes: 4

Dataset Information :

1. Class Name: 3 (L, B, R)
2. Left-Weight: 5 (1, 2, 3, 4, 5)
3. Left-Distance: 5 (1, 2, 3, 4, 5)
4. Right-Weight: 5 (1, 2, 3, 4, 5)
5. Right-Distance: 5 (1, 2, 3, 4, 5)

Ex - 3- Least Discriminant Analysisa(LDA)

Code - LDA:

""" Linear Discriminant Analysis is generally used in supervised learning algorithms as a classifier to separate the input data into two or more classes. It is also a dimensionality reduction technique. """

```
import numpy as np
from sklearn.datasets import load_iris

# ──────────────────────────────────────────────────
#  Load Iris dataset
# ──────────────────────────────────────────────────
iris = load_iris()
X = iris.data          # shape (150, 4)
y = iris.target        # 0,1,2
class_names = iris.target_names  # ['setosa', 'versicolor',
'virginica']

n_samples, n_features = X.shape
n_classes = len(np.unique(y))

print(f"Dataset: {n_samples} samples, {n_features} features,
{n_classes} classes\n")

# ──────────────────────────────────────────────────
# 1. Compute overall mean μ
# ──────────────────────────────────────────────────
mu = np.mean(X, axis=0)
print("Overall mean μ:")
print(np.round(mu, 4), "\n")

# ──────────────────────────────────────────────────
# 2. Class means & counts
# ──────────────────────────────────────────────────
means = np.zeros((n_classes, n_features))
counts = np.zeros(n_classes)

for c in range(n_classes):
    mask = (y == c)
    counts[c] = np.sum(mask)
    means[c] = np.mean(X[mask], axis=0)
    print(f"Class {c} ({class_names[c]:12})  n = {int(counts[c]):3d}
mean = {np.round(means[c], 4)}")

# ──────────────────────────────────────────────────
# 3. Between-class scatter SB
# ──────────────────────────────────────────────────
SB = np.zeros((n_features, n_features))
for c in range(n_classes):
    diff = (means[c] - mu).reshape(-1, 1)
    SB += counts[c] * (diff @ diff.T)
```

```python
print("\nBetween-class scatter SB:")
print(np.round(SB, 4))

# ────────────────────────────────────────────────
# 4. Within-class scatter SW
# ────────────────────────────────────────────────
SW = np.zeros((n_features, n_features))
for c in range(n_classes):
    mask = (y == c)
    diff = X[mask] - means[c]
    SW += diff.T @ diff

print("\nWithin-class scatter SW:")
print(np.round(SW, 4))

# ────────────────────────────────────────────────
# 5. Compute M = SW⁻¹ SB
# ────────────────────────────────────────────────
SW_inv = np.linalg.inv(SW)          # SW should be invertible for Iris
M = SW_inv @ SB

print("\nM = SW⁻¹ SB:")
print(np.round(M, 4))

# ────────────────────────────────────────────────
# 6. Very simple power iteration to find dominant eigenvector
# ────────────────────────────────────────────────
def simple_power_iteration(A, max_iter=100, tol=1e-9):
    n = A.shape[0]
    v = np.random.rand(n)
    v /= np.linalg.norm(v) + 1e-12

    for it in range(max_iter):
        v_new = A @ v
        eigenvalue = v @ v_new           # Rayleigh quotient
        v_new /= np.linalg.norm(v_new) + 1e-12

        if np.abs(np.dot(v, v_new) - 1.0) < tol:
            break
        v = v_new

    return eigenvalue, v, it + 1


# Find first (largest) eigenpair
lam1, v1, iter1 = simple_power_iteration(M)
v1 = v1 / np.linalg.norm(v1)          # ensure unit length
```

```
print(f"\n→ Largest eigenvalue ≈ {lam1:.6f}  (found in {iter1}
iterations)")
print("Eigenvector 1:", np.round(v1, 6))

# Deflate matrix to find second direction
M_deflated = M - lam1 * np.outer(v1, v1)

lam2, v2, iter2 = simple_power_iteration(M_deflated)
v2 = v2 / np.linalg.norm(v2)

print(f"→ Second eigenvalue ≈ {lam2:.6f}  (found in {iter2}
iterations)")
print("Eigenvector 2:", np.round(v2, 6))

# ────────────────────────────────────────────────
# 7. Projection matrix W (first two eigenvectors as columns)
# ────────────────────────────────────────────────
W = np.column_stack((v1, v2))

# Project all data to 2D
X_lda = X @ W

# ────────────────────────────────────────────────
# 8. Show some results
# ────────────────────────────────────────────────
print("\nFirst 6 projected points (should be mostly setosa):")
print(np.round(X_lda[:6], 4))

print("\nClass centers in the 2D LDA space:")
for c in range(n_classes):
    center = np.mean(X_lda[y == c], axis=0)
    print(f"{class_names[c]:12} : {np.round(center, 4)}")
```

Step - 0 Import the necessary libraries

```
import pandas as pd
import numpy as np
```

Step - 1 Load the dataset

```
url =
'https://archive.ics.uci.edu/ml/machine-learning-databases/balance-
scale/balance-scale.data'

data = pd.read_csv(url,header=None)# read url(The URL must be a string
with full slashes)
# url = " httpsarchive.ics.uci.edumlmachine-learning-databasesbalance-
scalebalance-scale.data"

url =
"https://archive.ics.uci.edu/ml/machine-learning-databases/balance-
```

```
scale/balance-scale.data"

# Declare the column names(This dataset does not have a header row, so
we define column names manually)
col_names = ['Class Name', 'Left-Weight', 'Left-Distance', 'Right-
Weight', 'Right-Distance']

#Read the data in the url using read_csv
data = pd.read_csv(url,header=None,names=col_names)

data = pd.DataFrame(data)
print(data.head())
```

```
  Class Name  Left-Weight  Left-Distance  Right-Weight  Right-Distance
0          B            1              1             1               1
1          R            1              1             1               2
2          R            1              1             1               3
3          R            1              1             1               4
4          R            1              1             1               5
```

Step 3 - Preprocess the Dataset

```
#Class name is categorical so we keep it aside
x = data.drop('Class Name', axis = 1)

#then we store it seperately as Y
Y = data['Class Name']

print(x.head())
```

```
   Left-Weight  Left-Distance  Right-Weight  Right-Distance
0            1              1             1               1
1            1              1             1               2
2            1              1             1               3
3            1              1             1               4
4            1              1             1               5
```

```
#Make sure you convert the integer values of X in to decimal values
X = x.to_numpy().astype(float)

# Map class labels to integers 0,1,2,... (L = left, B = balance, R =
right)
class_map = {name: i for i, name in enumerate(sorted(Y.unique()))}
y = Y.map(class_map).to_numpy()

print(X)
print("Mapping of class labels are :",class_map)
```

```
[[1. 1. 1. 1.]
 [1. 1. 1. 2.]
 [1. 1. 1. 3.]
```

```
 ...
 [5. 5. 5. 3.]
 [5. 5. 5. 4.]
 [5. 5. 5. 5.]]
Mapping of class labels are : {'B': 0, 'L': 1, 'R': 2}
```

```
class_names = sorted(class_map.keys())        # e.g. ['B', 'L', 'R']
n_samples, n_features = X.shape
n_classes = len(class_names)
print(f"Dataset: {n_samples} samples, {n_features} features,
{n_classes} classes")
print(f"Classes : {class_names}\n")
```

```
Dataset: 625 samples, 4 features, 3 classes
Classes : ['B', 'L', 'R']
```

Procedure

---

1. Overall mean μ

```
mu = np.mean(X,axis=0)
print("Overall mean μ:")
print(np.round(mu,4),"\n")
```

```
Overall mean μ:
[3. 3. 3. 3.]
```

---

2. Class means & counts

```
means = np.zeros((n_classes, n_features))
counts = np.zeros(n_classes)

for i, name in enumerate((n_classes, n_features)) :
    mask = (y==i)
    counts[i] = np.sum(mask)
    means[i] = np.mean(X[mask], axis=0) if counts[i] > 0 else
np.zeros(n_features)
    print(f"Class {i} ({name:2})   n = {int(counts[i]):4d}   mean =
{np.round(means[i], 4)}")
```

```
Class 0 ( 3)   n =   49   mean = [2.9388 2.9388 2.9388 2.9388]
Class 1 ( 4)   n =  288   mean = [3.6111 3.6111 2.3993 2.3993]
```

3. Between-class scatter SB

```
SB = np.zeros((n_features, n_features))
for i in range(n_classes) :
    diff  = (means[i] - mu).reshape(-1,1) # Calculates the difference
between the
                                          # class mean (mi) and the
global
                                          # mean (mu). It then
reshapes this
                                          # into a column vector (d *
1)
                                          # to prepare for matrix
multiplication.
    SB += counts[i] * (diff @ diff.T)     # diff @ diff.T - (Xi -
mu)*(Xi - mu)T performs an outer product, resulting in a d*d matrix.
                                          # counts[i] - Ni
print("\nBetween-class scatter SB:")
print(np.round(SB, 4))
```

```
Between-class scatter SB:
[[ 107.7392  107.7392 -105.5385 -105.5385]
 [ 107.7392  107.7392 -105.5385 -105.5385]
 [-105.5385 -105.5385  104.1038  104.1038]
 [-105.5385 -105.5385  104.1038  104.1038]]
```

4. Within-class scatter SB

```
SW = np.zeros((n_features, n_features))

for i in range(n_classes) :
    mask = (y==i)                     # Creates a boolean filter to
select only the
                                      # rows in X belonging to the
current class \(i\).
    if np.sum(mask) > 1 :
        diff = X[mask] - means[i]     # Centers the data for class \
(i\) by subtracting
                                      # the class mean from every
```

```
sample in that class.
        SW += diff.T @ diff
print("\nWithin-class scatter SW:")
print(np.round(SW, 4))
```

```
Within-class scatter SW:
[[2696.2608 1446.2608 2706.5385 2706.5385]
 [1446.2608 2696.2608 2706.5385 2706.5385]
 [2706.5385 2706.5385 4793.8962 3543.8962]
 [2706.5385 2706.5385 3543.8962 4793.8962]]
```

5. Compute M = SW^{-1} SB

Important: for the balance-scale dataset SW is often singular → use pseudo-inverse

```
try :
    SW_inv = np.linalg.inv(SW)
    print(" -> SW is invertible")
except np.linalg.LinAlgError :
    print("-> SW is singular -> using pseudo-inverse")
    SW_inv = np.linalg.pinv(SW)

M = SW_inv @ SB
print("\nM = SW⁻¹ SB (or pseudo-inverse):")
print(np.round(M, 4))
```

```
 -> SW is invertible

M = SW⁻¹ SB (or pseudo-inverse):
[[ 0.2806  0.2806 -0.2756 -0.2756]
 [ 0.2806  0.2806 -0.2756 -0.2756]
 [-0.1948 -0.1948  0.1914  0.1914]
 [-0.1948 -0.1948  0.1914  0.1914]]
```

6. Simple power iteration – find dominant eigenvectors

```
def simple_power_iteration(A,max_iter=200, tol =1e-10) :
    n = A.shape[0]
    v = np.random.rand(n)
    v /= np.linalg.norm(v) + 1e-12

    for it in range(max_iter):
        v_new = A @ v
```

```
        eigenvalue = v @ v_new
        v_new /= np.linalg.norm(v_new) + 1e-12

        if np.abs(np.dot(v, v_new) - 1.0) < tol:
            break
        v = v_new

    return eigenvalue, v, it + 1


# First direction (largest eigenvalue)
lam1, v1, iter1 = simple_power_iteration(M)
v1 = v1 / np.linalg.norm(v1)
print(f"\n→ Largest eigenvalue ≈ {lam1:.6f} (found in {iter1}
iterations)")
print("Direction 1:", np.round(v1, 6))

# Deflate and find second direction
M_deflated = M - lam1 * np.outer(v1, v1)
lam2, v2, iter2 = simple_power_iteration(M_deflated)
v2 = v2 / np.linalg.norm(v2)
print(f"→ Second eigenvalue ≈ {lam2:.6f} (found in {iter2}
iterations)")
print("Direction 2:", np.round(v2, 6))

# ──────────────────────────────────────────────
# 7. Projection matrix W (top-2 directions)
# ──────────────────────────────────────────────
W = np.column_stack((v1, v2))

# Project data to 2D LDA space
X_lda = X @ W

# ──────────────────────────────────────────────
# 8. Results summary
# ──────────────────────────────────────────────
print("\nClass centers in the 2D LDA space:")
for i, name in enumerate(class_names):
    center = np.mean(X_lda[y == i], axis=0)
    print(f"{name:6} : {np.round(center, 4)}")

# Optional: show a few projected points
print("\nFirst 8 projected points:")
print(np.round(X_lda[:8], 4))
```

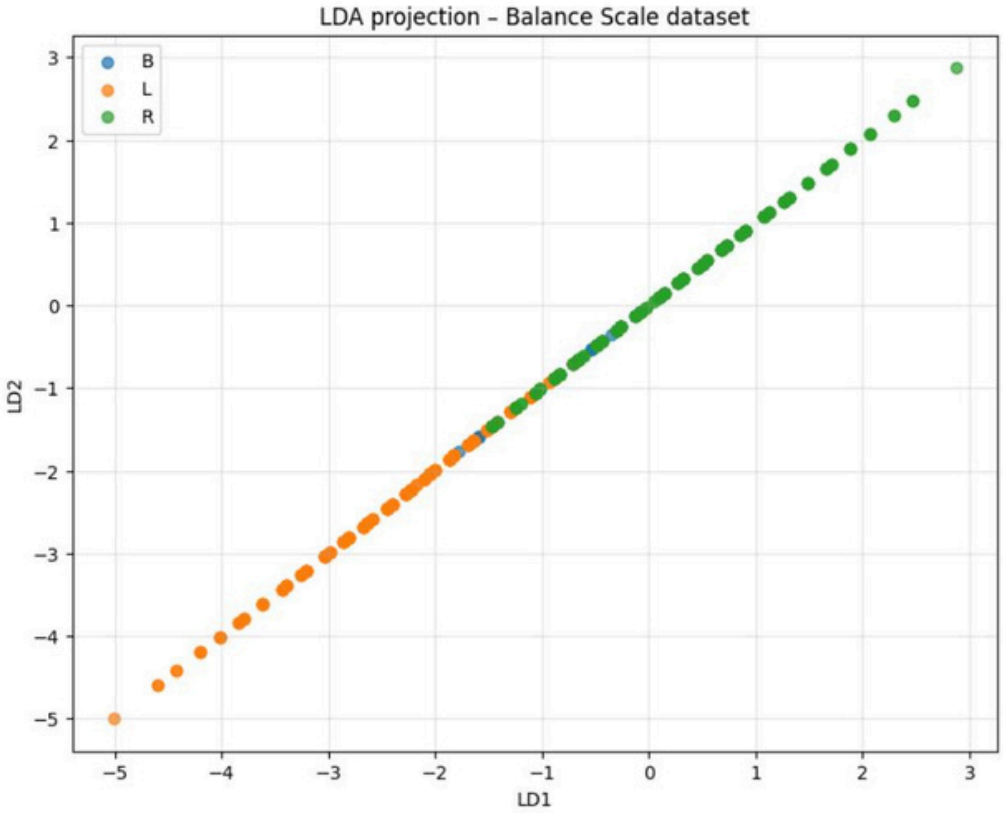
```
→ Largest eigenvalue ≈ 0.943846 (found in 3 iterations)
Direction 1: [-0.580804 -0.580804  0.40332   0.40332 ]
→ Second eigenvalue ≈ 0.000205 (found in 200 iterations)
Direction 2: [-0.58069  -0.58069   0.403484  0.403484]
```

```
Class centers in the 2D LDA space:
B      : [-1.0432 -1.0415]
L      : [-2.2593 -2.2577]
R      : [0.1258 0.1275]

First 8 projected points:
[[-0.355  -0.3544]
 [ 0.0484  0.0491]
 [ 0.4517  0.4526]
 [ 0.855   0.856 ]
 [ 1.2583  1.2595]
 [ 0.0484  0.0491]
 [ 0.4517  0.4526]
 [ 0.855   0.856 ]]
```

```
import matplotlib.pyplot as plt

plt.figure(figsize=(9,7))
for i, name in enumerate(class_names):
    mask = (y == i)
    plt.scatter(X_lda[mask,0], X_lda[mask,1], label=name, alpha=0.7)
plt.xlabel("LD1")
plt.ylabel("LD2")
plt.title("LDA projection — Balance Scale dataset")
plt.legend()
plt.grid(True, alpha=0.3)
plt.show()
```


LDA projection – Balance Scale dataset
B
L
R
LD2
LD1
3
2
1
0
−1
−2
−3
−4
−5
−5
−4
−3
−2
−1
0
1
2
3